

Visible Rows, Hidden Events: Certified Aggregates for Relation-Incomplete Event Streams

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Abstract

Many event streams are row-complete but relation-incomplete: every transaction is visible, yet the key joining rows into shared referrals, visits, service episodes, or vehicle runs is suppressed. Choosing one linkage can turn this structural uncertainty into apparently precise but decision-unstable statistics. We introduce EVENTFRONTIER, which optimizes aggregate queries over all temporally ordered event decompositions consistent with released rows, timestamp supports, and a declared simultaneous-capacity bound. An event need not be a clique: sequential members may form an A - B - C overlap chain although A and C never coexist. We prove joint nesting under timestamp support expansion and capacity relaxation. For a fixed root and span, connectivity and occupancy have a consecutive-ones formulation, yielding an integral LP pricing oracle. Dantzig-Wolfe decomposition closes the full LP; branching on row usage and Ryan-Foster co-membership, while reusing exact branch-compatible pricing, gives a certified integer algorithm on medium instances. In 3,000 controlled-truth experiments, full candidate support covers the true aggregate in every instance, while a feasible temporal point rule makes threshold errors in 17.4–19.0% of cases; every such error is flagged by a frontier that crosses the threshold. Retaining about 84% of true members leaves the complete true event world representable only 31–33% of the time. Public Chicago and NYC audits then separate hidden-linkage, timestamp-support, and capacity ambiguity. These are conditional feasible-world claims: we do not recover operational event identities, realized capacity, or population effects.

CCS Concepts

• **Information systems** → **Data provenance**; **Data integration**;
• **Theory of computation** → **Mathematical optimization**; •
Computing methodologies → *Structured outputs*.

Keywords

partial identification, relation-incomplete data, hidden event membership, interval graphs, branch-and-price

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1 Introduction

Many data releases preserve rows while deleting relationships. A stream may report each encounter's time, location, duration, and outcome but suppress the key that says which rows belonged to the same referral episode, household visit, field-service tour, shared ride, or other bounded-capacity event. Analysts can still count transactions, but event-level composition, exposure, dispersion, utilization,

and threshold decisions depend on an unobserved partition. A clustering or linkage algorithm returns one plausible partition; it does not by itself establish that the downstream conclusion is invariant to the other partitions that remain compatible with the release.

We call such streams *row-complete and relation-incomplete*. Their central knowledge-discovery problem is not always to guess the missing relation. Sometimes the scientifically relevant question is instead:

Which aggregate conclusions survive every latent event world allowed by the released information and explicitly stated structural assumptions?

This distinction matters whenever a point reconstruction is used to trigger a policy threshold, rank subpopulations, or compare operating regimes. A point estimate may be feasible and reproducible yet still sit on the wrong side of a decision boundary relative to another equally feasible world.

Ride pooling provides a demanding public audit instance, but not the scope of the framework. A released row may state that a trip was successfully shared without exposing a co-rider or vehicle-run key. High-capacity events are not pairwise cliques: A may overlap B , B may overlap C , and A may never overlap C , while all three belong to one sequential run whose simultaneous occupancy is two. The same ordered-event structure appears when cases, customers, or visits enter and leave a shared episode over time. Timestamp coarsening creates a second ambiguity. Treating the full public envelope as the realized interval is generally wrong because it can simultaneously create artificial overlap and artificial congestion. The correct partial-identification quantifier asks whether *there exists* a latent exact timestamp completion inside every released support.

We introduce EVENTFRONTIER, a certification framework for aggregate queries on such event streams. A visible candidate universe is divided into required core rows and optional boundary buffers. A feasible world chooses exact intervals inside released timestamp supports and partitions selected rows into connected, bounded-occupancy temporal events. Rather than outputting a preferred graph, the method minimizes and maximizes a downstream functional over all feasible worlds. Every displayed endpoint is accompanied by an attained witness internally or by an explicit solver status; unresolved computations are not converted into infeasibility. The resulting interval is conditional on the declared candidate universe, supports, connectivity rule, and capacity. It is not a frequentist confidence interval, nor does it claim that every interior scalar is attainable.

This paper makes four contributions.

- (1) **Identification geometry.** We formalize core-buffer ordered-event worlds with set-valued timestamps, positive-overlap connectivity, occupancy capacity, and aggregate functionals. We prove joint nesting under timestamp support expansion and capacity relaxation and define a capacity-equivalent information-loss index.

- (2) **Structure-exploiting optimization.** For one fixed-root, fixed-span event, interleaved segment-coverage and boundary-bridge rows have the consecutive-ones property. This gives an integral LP oracle for additive event pricing and an exact Dantzig–Wolfe certificate for the full master relaxation.
- (3) **Certified integer decomposition.** The cross-event master is not integral. We therefore branch first on optional-row usage and then on Ryan–Foster pair co-membership. Each branch is priced through a finite family of forced-in/forced-out versions of the same interval LP. Exhausting the branch queue certifies the global integer support optimum; timeouts retain valid lower and upper bounds.
- (4) **Validation and public audits.** In 3,000 controlled-truth instances, we evaluate point reconstructions, threshold decisions, and candidate truncation separately from city evidence. Chicago and NYC then expose complementary release regimes and separate hidden-linkage, timestamp-support, and capacity ambiguity without publishing reconstructed links.

The controlled study shows why certification changes the knowledge-discovery answer. Conditional on the true support count but not membership labels, a feasible temporal point rule makes the wrong decision at predeclared thresholds in 17.4–19.0% of cases across capacities two through four. Every error occurs when the certified frontier crosses the threshold, so EVENTFRONTIER correctly refuses to present that decision as relation-invariant. Candidate completeness is a distinct Gate: keeping roughly 84% of true members leaves the full true event world representable only 31–33% of the time, although a nonempty frontier at the true support count usually still exists. Row-level recall can therefore substantially overstate relational coverage.

The public audits provide complementary geometry rather than recovery accuracy. In a fixed NYC pairwise benchmark, artificial 15-minute time coarsening raises candidate edges by 79.5% and several identified widths by 12–15%. In an ordered-event common-support comparison, most of the $C = 2$ to $C = 4$ widening occurs at the first capacity relaxation. Across six predeclared audit windows, artificial timestamp supports increase certified reachable support at both $C = 2$ and $C = 3$ in every window, with regime-dependent magnitude. On a small exact-time public cohort, column generation needs 102 of 1,719 feasible capacity-four event columns to close the LP, and branch-and-price independently certifies the integer optimum. Fine temporal resolution therefore does not imply fine relational resolution, while tractable local event structure does not imply an integral global partition.

2 Related Work

Uncertain linkage and aggregate queries. Record-linkage research represents globally compatible links as constrained matchings rather than independent pair decisions [14]. Database work propagates linkage uncertainty to aggregate queries [10] and optimizes extremal aggregate answers over candidate-constrained two-table integrations [19]. Attribute-uncertain databases likewise track value bounds through relational queries [7]. We share the possible-worlds principle but study a different latent object: rows inside one event stream must be partitioned into temporally ordered groups whose total membership may exceed simultaneous capacity, while exact

timestamps may themselves be set-valued. The estimand is an aggregate frontier over joint grouping and time completions, not a posterior over pair identities.

Temporal grouping and mobility applications. Shareability networks and dynamic trip–vehicle assignment solve a forward allocation problem when requests and candidate trips are observed [1, 13, 15]. Empirical work predicts sharing or matching outcomes [12, 18], studies pooled demand across neighborhoods [17], or constructs proxy groups by spatiotemporal clustering [16]. Our city audit is inverse: the release preserves transaction rows and outcomes but removes the event key. More broadly, the same inferential object arises in referral, household-visit, and service-tour streams. We do not select the most likely cluster; we certify which downstream aggregate claims survive all declared temporal-event decompositions.

Structured optimization. Integer programming is standard for globally coupled disclosure constraints [8], and bounded-width dynamic programming is standard in structured inference [6, 9]. Our decomposition exploits a more specific interval structure. One rooted event with a fixed span has an integral consecutive-ones formulation, which becomes an exact pricing oracle for a Dantzig–Wolfe master. The master itself is nonintegral, so we use branch-compatible row-usage and pair-co-membership disjunctions. This separates a polynomial local pricing problem from a potentially exponential global integer search while retaining valid anytime bounds.

Candidate restrictions and calibrated learning. Learning can sharpen a feasible-world set, but node-level event indicators do not identify partner probabilities without assumptions. Matching-level conformal prediction and weakly supervised structured prediction have well-developed foundations [2, 11]. We keep learned scoring outside the central contribution. Headline candidate rules are score-free and outcome-blind; the controlled study instead asks whether simple point reconstructions yield decisions invariant to the certified frontier and how candidate truncation affects full-world coverage.

3 Latent Ordered-Event Worlds

Released rows and boundary roles. Let $V = C_0 \cup B_0$ be a finite candidate universe. Core rows C_0 define the target cohort. Buffer rows B_0 lie near its temporal boundary and may belong to an event containing a core row, but need not be selected. This distinction prevents an arbitrary observation window from forcing every nearby row into the target population.

For row i , the release supplies a timestamp support

$$\mathcal{T}_i \subseteq \{(s, e) : s < e\}.$$

Exact public time is the singleton case. A latent completion chooses $t_i = (s_i, e_i) \in \mathcal{T}_i$ for every row. We use half-open intervals $I_i = [s_i, e_i)$ and define positive overlap by $|I_i \cap I_j| > 0$.

Ordered temporal events. For a completed timestamp vector t , a subset $R \subseteq V$ is a feasible event at simultaneous capacity C if: (i) R contains at least two rows and at least one core; (ii) the graph on R joining positive-overlap pairs is connected; and (iii)

$$\max_u \sum_{i \in R} \mathbb{I}\{u \in I_i\} \leq C. \quad (1)$$

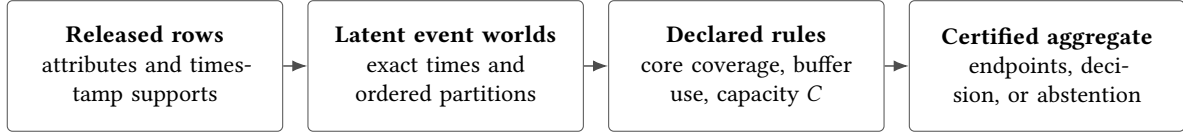


Figure 1: EVENTFRONTIER does not output a preferred missing relation. The release and analyst-declared support rules define latent event worlds; optimization discovers the aggregate statements invariant across those worlds.

Condition (1) limits simultaneous occupancy, not total event cardinality. Thus one event may contain more than C rows through sequential turnover. In the ride-pooling audit such an event is a latent vehicle run; in other streams it may be a referral episode, household visit, or service tour.

A world $F = (t, \Pi)$ consists of a timestamp completion and a collection of disjoint feasible events such that every core row belongs to exactly one event and every buffer row belongs to at most one. Unselected buffers remain outside Π . Let $q(F)$ denote the number of selected buffers. For fixed q , write

$$\mathcal{F}(\mathcal{T}, C, q) = \{F : q(F) = q\}.$$

The reachable support set and maximum are

$$Q(\mathcal{T}, C) = \{q : \mathcal{F}(\mathcal{T}, C, q) \neq \emptyset\}, \quad M(\mathcal{T}, C) = \max Q(\mathcal{T}, C).$$

Aggregate functionals. Let $H(F)$ be any event-label-invariant aggregate computable from public row attributes and the latent partition, such as mean outcome among selected buffers, within-event dispersion, exposure, or event count per core. For nonempty $\mathcal{F}(\mathcal{T}, C, q)$ define

$$\underline{H}(\mathcal{T}, C, q) = \min_{F \in \mathcal{F}(\mathcal{T}, C, q)} H(F), \quad \bar{H}(\mathcal{T}, C, q) = \max_{F \in \mathcal{F}(\mathcal{T}, C, q)} H(F). \quad (2)$$

Both extrema are attained for the finite fixed-time models used below. The continuous-support implementation reports a feasible witness, certified infeasibility, or unresolved status rather than coercing a timeout.

PROPOSITION 1 (SUPPORT AND CAPACITY NESTING). *Suppose $\mathcal{T}_i \subseteq \mathcal{T}_i'$ for every row and $C \leq C'$. Then, for every q ,*

$$\mathcal{F}(\mathcal{T}, C, q) \subseteq \mathcal{F}(\mathcal{T}', C', q).$$

Consequently $Q(\mathcal{T}, C) \subseteq Q(\mathcal{T}', C')$ and $M(\mathcal{T}, C) \leq M(\mathcal{T}', C')$. For fixed q and any H ,

$$\underline{H}(\mathcal{T}', C', q) \leq \underline{H}(\mathcal{T}, C, q), \quad \bar{H}(\mathcal{T}', C', q) \geq \bar{H}(\mathcal{T}, C, q).$$

The fixed- q condition matters. Comparing outcomes at each capacity's own maximum $M(\mathcal{T}, C)$ changes both the world set and the conditioning event; those ranges need not widen monotonically.

Capacity-equivalent information loss. For fine support $\mathcal{T}$ and coarser support $\mathcal{T}' \supseteq \mathcal{T}$, define

$$\kappa_{\mathcal{T} \rightarrow \mathcal{T}'}(C) = \min\{k \geq 0 : M(\mathcal{T}, C + k) \geq M(\mathcal{T}', C)\}. \quad (3)$$

This measures how many fine-time capacity steps are required to reach the support attainable under coarse time. When a coarse solver only certifies a feasible lower bound $L(\mathcal{T}', C) \leq M(\mathcal{T}', C)$, replacing M by L yields a conservative lower bound on the comparison, not a sharp equality.

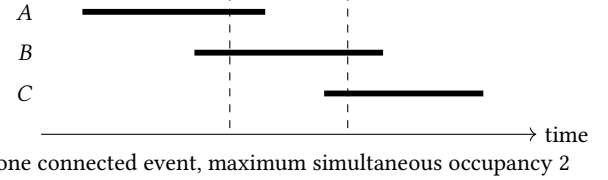


Figure 2: An ordered event need not be a clique. A overlaps B, B overlaps C, and A does not overlap C. The event has three members but capacity two.

4 Structure-Exploiting Optimization

4.1 The declared $K = 2$ special case

If every event must contain exactly two rows, the world reduces to a core-cover matching: every core has degree one, every buffer has degree at most one, and buffer–buffer edges are excluded. Pair-additive queries become weighted matching objectives. This is the Chicago benchmark and a useful lower-complexity special case, but it cannot represent sequential higher-cardinality events such as Figure 2.

4.2 A compact connectivity characterization

Fix exact intervals, a root row r , and a candidate span $[a, b]$. Let the sorted distinct endpoints inside the span induce elementary open segments. For a selected set containing r , positive-overlap connectivity with union span $[a, b]$ is equivalent to:

- (1) every elementary segment is covered by at least one selected interval;
- (2) every internal endpoint boundary is strictly crossed by at least one selected interval.

The second condition excludes components that only touch at an endpoint. Capacity adds an upper bound of C to each segment-coverage row.

Interleave segment and boundary rows in temporal order. Each interval column then contains ones in one consecutive block. This yields the pricing primitive.

THEOREM 1 (INTEGRAL ROOTED SINGLE-EVENT ORACLE). *For fixed exact intervals, root r , span $[a, b]$, capacity C , and additive row weights, the linear relaxation of the segment-coverage, boundary-bridge, capacity, root, and forced-companion formulation has an integral optimum. Hence a minimum- or maximum-weight rooted event can be found by linear programming. Enumerating observed endpoint spans and, only when needed, one forced companion gives a polynomial-time exact single-event oracle.*

Proof sketch. The interleaved segment/boundary incidence matrix has the consecutive-ones property by columns. Signed duplicate rows for lower and upper coverage bounds, plus identity rows fixing the root and companion, preserve total unimodularity. The right-hand side is integral, so every extreme point is integral. Candidate spans need only use observed interval endpoints. Appendix details and exhaustive small-library comparisons are included in the artifact.

4.3 Dantzig–Wolfe decomposition

Let $\mathcal{R}_C$ be all feasible exact-time event columns. For event R , let $b_R = |R \cap B_0|$ and introduce $\lambda_R \geq 0$. The full relaxation for maximum reachable buffer support is

$$\max_{\lambda} \sum_{R \in \mathcal{R}_C} b_R \lambda_R \quad (4)$$

$$\text{s.t.} \quad \sum_{R \ni i} \lambda_R = 1 \quad i \in C_0, \quad (5)$$

$$\sum_{R \ni j} \lambda_R \leq 1 \quad j \in B_0. \quad (6)$$

Core equalities and buffer inequalities couple otherwise independent event columns. Given master duals, reduced cost is additive across selected rows, so Theorem 1 gives exact pricing. A phase-one artificial-core master establishes feasibility.

PROPOSITION 2 (FULL-MASTER LP CERTIFICATE). *If every rooted pricing call is solved exactly and the terminal minimum reduced cost is nonnegative, the current restricted-master value equals the optimum of Eqs. (4)–(6) over all feasible event columns.*

Local integrality does not transfer to the coupled master. The appendix gives a capacity-two instance with

$$\text{OPT}_{LP} = 4 > 3 = \text{OPT}_{IP}.$$

A restricted integer solve over columns generated only for the root LP can therefore miss the full integer optimum.

4.4 Branch-compatible exact integer decomposition

We close the support-maximization integer master with branch-and-price. At a node, optional buffers may be fixed to total usage zero or one. Once row usage is integral but event columns remain fractional, Ryan–Foster branching chooses two active rows and creates:

together: both rows have identical membership in every event,

For a priced root, together and separate restrictions are expanded into a finite set of forced-in/forced-out cases. Each case adds only identity fixes and variable bounds to the fixed-span formulation, so Theorem 1 still solves it exactly. The number of cases may grow exponentially with branch depth; we make no polynomial-time claim for the full integer problem.

PROPOSITION 3 (INTEGER CERTIFICATE). *Suppose each node relaxation is closed by exact branch-compatible pricing, every child exhaustively implements its row-usage or Ryan–Foster disjunction, and the branch queue terminates. Then the incumbent is a globally*

Table 1: Controlled truth over 3,000 relation-incomplete event instances. Decision error is averaged over three predeclared thresholds.

C	Instances	Median width	Point MAE	Decision error
2	1,000	0.446	0.121	17.9%
3	1,000	0.459	0.118	17.4%
4	1,000	0.473	0.125	19.0%

optimal integer solution of the full event-column master. If computation stops early, the best incumbent and largest open-node LP bound form a valid global lower and upper bound.

Justification. The two children partition the integer solutions compatible with the parent. Exact pricing makes each node bound valid for the full, not merely restricted, column set. Standard branch-and-bound dominance then proves the claim. The implementation replays every incumbent, records open-node bounds, and never labels a time-limited cell optimal.

4.5 Existential timestamp completion

For set-valued timestamps, a world is feasible when there *exists* one exact interval choice inside each support that satisfies connectivity and capacity. Replacing a support by its entire outer envelope is neither a valid inner nor outer model for ordered events: it can add bridges while also creating artificial concurrency. Our continuous mixed-integer model therefore chooses latent starts and ends jointly with event membership. Each requested support count receives one of four statuses: certified feasible witness, proven infeasible, unresolved without incumbent, or unresolved invalid incumbent. Only the first two are treated as resolved.

5 Controlled-Truth Validation

The public city releases do not expose event identifiers, so they cannot measure relation-recovery accuracy. We therefore separate method validation from public auditing through a controlled generator with known ordered-event membership. Each instance contains three required cores, eight candidate buffers, public time intervals, and a public scalar outcome. Every core has a direct member and one or two events receive a sequential member that can connect through the direct member without overlapping the core. Remaining buffers are temporally plausible distractors. Membership labels generate and evaluate the instance but are never supplied to EVENTFRONTIER or the point rules.

We never provide seeds at each $C \in \{2, 3, 4\}$. The composition task is conditioned on the true selected-buffer count q , while hiding membership, to isolate relation ambiguity from support-count error. The point comparator chooses, among feasible worlds at q , the world maximizing an outcome-blind temporal-proximity score. Three fixed thresholds, 0.25, 0.50, and 0.75, convert the mean selected-buffer outcome into a downstream decision. A decision is certified only when the full frontier lies on one side of the threshold.

With the full candidate universe, the true aggregate lies in the certified frontier in all 3,000 instances. The feasible temporal point is often close in absolute error yet makes a threshold decision error in roughly one of six comparisons (Table 1). Every such error occurs

Table 2: Candidate truncation after retaining six of eight buffers.

C	Member recall	World repr.	Frontier exists	Scalar covered
2	83.4%	31.4%	98.9%	69.2%
3	83.8%	31.9%	100.0%	71.1%
4	83.8%	32.6%	100.0%	71.7%

when the frontier straddles the threshold:

$$\Pr(\text{frontier flags ambiguity} \mid \text{point decision is wrong}) = 1.$$

Thus the method’s downstream value is not to promise a superior missing-link guess. It identifies when the desired conclusion is invariant to all admissible relations and abstains precisely on these observed point-rule failures.

Candidate recall is not world coverage. We next rank buffers by the same outcome-blind temporal score and retain only four, six, or all eight. Table 2 reports the intermediate six-buffer screen. Mean true-member recall remains about 84%, but the entire true relational world is representable only 31–33% of the time. Nevertheless, a nonempty frontier at the true support count exists in 98.9–100% of instances. Feasibility of the optimization problem is therefore not evidence that the candidate universe still contains truth.

Scalar coverage can also survive by coincidence after the joint world is removed: this occurs in 37.8–39.2% of instances. Consequently, validating only an aggregate, or only average candidate recall, can conceal a misspecified relation universe. The correct workflow reports candidate-support sensitivity as a separate identification layer. Pair matching and connected components are also evaluated in the artifact, but their failure rates are generator-specific: the generator intentionally includes sequential non-clique events and bounded occupancy. We therefore use them as diagnostic misspecification baselines, not as estimates of real-world algorithm accuracy.

6 Public-Data Audit Design

Complementary releases. Chicago’s TNP release provides a declared two-transaction shared-trip benchmark with nearest-15-minute public times and privacy-coarsened geography [3–5]. NYC’s High Volume For-Hire Vehicle release provides second-level pickup and drop-off times, shared-match indicators, Taxi Zones, trip miles, and trip time, but no public co-rider or run key. We use the official 2023 Open Data table¹ for audited exact optimization and the May 2026 monthly Parquet only for current candidate-multiplicity scale. These datasets instantiate a general relation-incomplete event problem; neither supplies ground truth for operational event membership.

Visible candidate universes. A candidate rule is part of the estimand rather than hidden preprocessing. Chicago uses a boundary-complete core-incident temporal universe and reports radius and omitted-incidence sensitivity separately. NYC uses provider and temporal proximity; exact Taxi Zone equality is only a sensitivity

screen because it can delete plausible relations. No learned score or public outcome enters the headline candidate construction.

Certification levels. The experiments are certification audits, not population estimation.

- (1) A current-month NYC count audit measures shared-match rows and candidate multiplicity for a fixed 200-row cohort.
- (2) A fixed 2023 pairwise cohort contains 38 cores and 399 buffers and compares exact seconds with artificial 15-minute outer intervals.
- (3) Ordered-event audits use exact or set-valued time and $C \in \{2, 3, 4\}$. A common-support cohort has eight cores; a separately predeclared replication panel uses four cores and 12 outcome-blind temporal buffers in each of six windows.
- (4) A four-core, 12-buffer exact-time cohort compares complete event-column enumeration, column generation, and exact branch-and-price. The same cohort is used only as a small algorithmic audit, not as a city-scale claim.

The six panel windows were frozen before their frontiers were inspected. They span weekday morning, weekday evening, weekend evening, and three seasons, but remain purposive rather than probabilistically sampled.

Solver and disclosure contracts. Exact fixed-time support frontiers use complete enumeration where declared. Column generation closes a full LP only after every rooted pricing problem is solved and no improving column remains. Branch-and-price calls an integer optimum only after the node queue closes; otherwise it retains an incumbent and open-node upper bound. Continuous existential-time cells use solver certificates and witness replay, and unresolved counts are never classified as infeasible. Artifacts publish aggregate tables, hashes, and statuses, but no raw identifiers, event columns, partner assignments, or latent timestamps.

7 Public-Data Results

7.1 Fine timestamps do not identify the hidden relation

The May 2026 NYC file contains 222,149 public shared-match rows. In a fixed 200-row cohort, exact public interval overlap yields 154,708 directed core-candidate incidences over 1,042 candidate nodes; the median core degree is 786. Requiring equal pickup and drop-off Taxi Zones leaves only 44 incidences. Zone equality is therefore an aggressive candidate screen, not a defensible necessary condition for hidden co-membership.

Table 3 uses the separate fixed 2023 pairwise cohort. Exact seconds already leave broad ranges. Artificial 15-minute time coarsening adds 79.5% more candidate edges and increases the three displayed identified widths by 12.4–15.3%. Time coarsening aggravates ambiguity but does not create it.

7.2 Capacity uncertainty is front-loaded

To isolate the pure effect of capacity, Table 4 fixes the same 72 selected buffers across $C = 2, 3, 4$. Proposition 1 then guarantees nested feasible sets. Miles width nearly doubles from $C = 2$ to 3 but rises only 6.0% from $C = 3$ to 4; duration behaves similarly. In this cohort, 89.2% of the total miles widening and 86.2% of the

¹Dataset u253-aew4: <https://data.cityofnewyork.us/d/u253-aew4>.

Table 3: NYC pairwise benchmark: exact seconds versus artificial 15-minute outer intervals. Endpoint pairs are solver-certified with zero reported gap.

Quantity	Exact	15-min	Change
Candidate edges	5,837	10,475	+79.5%
Minimum core degree	130	260	+100.0%
Miles-gap width	12.285	13.807	+12.4%
Trip-time-gap width (min)	38.113	43.934	+15.3%
Same-dropoff-zone width	0.737	0.842	+14.3%

Table 4: Ordered-event outcome frontiers at a common support of 72 selected buffers.

C	Miles range	Width	Minutes range	Width
2	3.440–9.679	6.239	17.779–37.147	19.368
3	3.131–15.448	12.317	15.983–55.712	39.729
4	3.116–16.166	13.050	15.924–58.906	42.982

Table 5: Predeclared NYC existential-time panel. Each entry is exact maximum / coarse certified maximum / certified gain lower bound.

Window	$C = 2$	$C = 3$
January weekday AM	5/8/+3	9/12/+3
January weekday PM	4/8/+4	8/12/+4
January weekend PM	4/11/+7	8/12/+4
April weekday PM	7/8/+1	11/12/+1
July weekday PM	6/8/+2	10/12/+2
October weekday PM	5/8/+3	9/12/+3

duration widening from $C = 2$ to 4 occurs at the first relaxation. This is a cohort-specific identification result, not an estimate of realized physical capacity.

A different conditioning event explains an initially counterintuitive pattern. At each capacity’s own maximum support, composition ranges can contract because larger C forces inclusion of more rows rather than permitting an extremal subset. Holding support fixed restores the monotone set comparison. This separates the extensive margin—which rows can enter some feasible world—from the conditional composition margin.

7.3 Timestamp support and capacity are partially substitutable

Table 5 reports the six-window existential-time panel. Every window has a positive certified coarse-support gain at both $C = 2$ and $C = 3$. The $C = 2$ gain ranges from one to seven buffers. Artificial coarse $C = 2$ reaches the exact $C = 3$ maximum in two of six windows, whereas coarse $C = 3$ reaches the exact $C = 4$ maximum in all six. The latter comparison is capped by the 12-buffer design, so it indicates support substitution within the declared panel, not universal saturation.

The defensible conclusion is not that a 15-minute support is always “one extra seat.” Timestamp information and simultaneous capacity are partially substitutable dimensions of the identified set,

Table 6: Exact-time decomposition on four cores and 12 buffers. Runtime is for branch-and-price; full column counts are available only through this small enumeration audit.

C	Full cols.	LP cols.	LP share	LP/IP	Nodes	Seconds
2	54	54	100.0%	4/4	1	0.74
3	394	82	20.8%	8/8	4	10.47
4	1,719	102	5.9%	12/12	2	9.31

with a magnitude determined by local interval density and chain structure.

7.4 Exact decomposition needs few LP columns and closes the IP

Table 6 uses one exact-time public cohort. At $C = 4$, complete enumeration contains 1,719 feasible event columns, while 102 columns (5.9%) close the full LP. Solving the integer problem requires columns that may not be generated for the root LP alone: the restricted integer objectives at $C = 3, 4$ are seven and ten, below the full optima eight and 12. The branch-and-price layer closes both values while preserving exact pricing at every node.

The public cohort has essentially zero root integrality gap, so it does not by itself show why branching is necessary. The locked nonintegrality witness does: its root LP is four, its integer optimum is three, and branch-and-price closes the value after 17 nodes. Together, the two audits distinguish column economy from integer difficulty. They establish an exact medium-instance algorithmic path, not a city-scale runtime guarantee.

8 Discussion

Knowledge discovery under missing relations. The method changes the target from “recover one plausible relation” to “certify which downstream statements are invariant to the admissible relations.” The controlled experiments show why this is not a semantic distinction: a fully feasible point reconstruction can cross the wrong policy threshold, while the frontier correctly marks the conclusion as ambiguous. Conversely, when the entire frontier lies on one side, every feasible world agrees. This certified-decision view is portable to referral episodes, household or field visits, service tours, and other row-complete streams with hidden temporal grouping.

What is identified. Endpoints are sharp or certified only relative to the declared candidate universe, timestamp supports, connectivity rule, capacity, and query. They do not identify a unique partition. Expanding timestamp support or capacity cannot shrink a fixed-support feasible set, but comparisons at capacity-specific maximum support change the conditioning event. Candidate truncation is more fundamental: if a true event is deleted before optimization, endpoint certification cannot restore it. The controlled benchmark demonstrates that high member recall, a nonempty frontier, and even scalar coverage can coexist with omission of the true joint world.

Algorithmic boundary. Theorem 1 makes fixed-span event pricing polynomial, and Proposition 2 certifies the full LP without enumerating all columns. Proposition 3 supplies an exact integer

path through branch-compatible row and pair disjunctions. The full algorithm is not claimed polynomial: the number of forced-in/forced-out pricing cases can grow exponentially with branch depth. The public 4+12 audit is an independent correctness check, not evidence of city-scale closure. The next scaling question is empirical—how root gap, branching, and generated columns evolve with temporal density and core size—and can be reported honestly through anytime global gaps when optimality is not reached.

Baselines and external validity. Our point rules are deliberately simple and interpretable. They establish that feasibility or temporal plausibility does not guarantee a relation-invariant decision, but they are not a final contest against modern learned linkage. A stronger application study would train a scoring model on a separate ground-truth source, calibrate candidate recall without using target outcomes, and compare its point answer with the certified frontier. Public Chicago and NYC data cannot supply that operational ground truth. Their role is to audit the ambiguity geometry induced by real release operators, not to estimate partner-recovery accuracy.

9 Limitations and Ethical Considerations

Neither city release exposes a public vehicle-run key. We therefore do not claim actual co-riders, partner recall outside a declared candidate universe, production matching logic, realized pool size, or physical vehicle capacity. The artificial 15-minute NYC supports are an information-loss intervention, not an assertion about the TLC observation operator. The six-window panel is purposive and too small for a citywide prevalence or causal estimate. The controlled generator validates method logic under known truth but does not turn its error rates into operational accuracy estimates.

Numerical claims retain their certificate class. Complete enumeration is used only where explicitly stated; branch-and-price optima require an exhausted node queue; time-limited runs keep valid incumbent and open-node bounds. Candidate support remains an analyst-facing assumption rather than a hidden modeling choice.

The unit of analysis is a released transaction, not a stable person. Zones and neighborhood attributes describe context, not individual socioeconomic status or preferences. Extremal event assignments can create reidentification risk, so the public artifact suppresses row identifiers, event columns, and latent world witnesses and reports aggregate results and provenance metadata only.

10 Generative AI Usage

Generative AI assisted drafting, code review, counterexample search, and editing. All mathematical claims, empirical outputs, citations, and artifact states remain author-verified; generated text is not treated as city evidence.

11 Conclusion

Row-complete data can remain relation-incomplete enough to reverse downstream decisions. EVENTFRONTIER represents every admissible ordered temporal event, then reports only aggregate conclusions that survive that uncertainty. The identification geometry separates candidate support, timestamp support, and capacity. The optimization geometry separates an integral local pricing problem, a nonintegral global master, and an exact branch-and-price certificate. Across 3,000 controlled instances, plausible feasible point

answers make systematic threshold errors that the frontier consistently flags as ambiguous; candidate truncation further shows that row recall is not relational-world coverage. Chicago and NYC audits demonstrate that these issues persist in public event streams with complementary release regimes. The resulting object is not a recovered partner graph, but a certified boundary between discoverable and relation-dependent aggregate knowledge.

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A Proof of Proposition 1

Take $F = (t, \Pi) \in \mathcal{F}(\mathcal{T}, C, q)$. Since each $t_i \in \mathcal{T}_i \subseteq \mathcal{T}'_i$, the same completion is allowed by $\mathcal{T}'$. Every event in Π has the same positive-overlap graph under the unchanged completion, and its maximum occupancy is at most $C \leq C'$. Core and buffer incidence constraints and the selected-buffer count q are unchanged. Hence $F \in \mathcal{F}(\mathcal{T}', C', q)$. Set inclusion gives nesting of reachable counts and maxima. Minimizing a fixed functional over a superset can only weakly lower its optimum, and maximizing can only weakly raise it.

B Fixed-Span Connectivity and Total Unimodularity

Let S be a selected interval family with union span $[a, b]$. If the positive-overlap graph is connected, the union is connected and covers every elementary segment. At an internal endpoint boundary, a path from an interval active immediately to its left to one active immediately to its right must contain an interval crossing the boundary strictly; endpoint touching alone is not a positive-overlap edge.

Conversely, suppose every elementary segment is covered and every internal boundary is strictly crossed. Moving from left to right, the crossing interval at each boundary overlaps intervals covering the adjacent segments with positive length. Induction joins the segment-covering intervals into one positive-overlap component. Any selected interval lies inside the covered span and overlaps that component on a segment it covers, so the full family is connected.

Interleaving segment and boundary rows makes every interval column consecutive. Lower-coverage, upper-capacity, and bridge constraints are signed copies of this interval matrix; root and companion fixes are identity rows. The matrix is totally unimodular with integral right-hand side, proving Theorem 1.

C Proof Detail for Proposition 3

At every branch node, exact pricing establishes the optimum of the node LP over the full column universe compatible with that node. If a free optional row has fractional total usage $z_i \in (0, 1)$, every integer solution satisfies either $z_i = 0$ or $z_i = 1$, so the two children partition the parent integer set.

When all row usages are integral but the column vector remains fractional, Ryan–Foster branching selects active rows i, j with fractional co-membership

$$0 < \sum_{R \ni i, j} \lambda_R < 1.$$

Every integer partition places i and j either in the same event or in different events. The together and separate children again form an exhaustive disjunction. In a priced event column, together means either both rows are selected or both are excluded; separate means at least one is excluded. Enumerating these finite Boolean cases yields forced-in/forced-out subproblems. For each root and span, they add only variable bounds and identity equalities to the totally unimodular interval formulation, so pricing remains exact.

Node LP values are valid upper bounds for support maximization. Incumbents are valid lower bounds after replay against core-equality, buffer-at-most-once, connectivity, and capacity constraints.

Standard best-bound branch-and-bound therefore preserves the global gap. When the queue is empty, no unexplored integer solution can exceed the incumbent; if the queue remains nonempty, the largest open-node LP value remains a valid global upper bound.

D Master Nonintegrality Witness

At capacity two, use four cores $c_0 = [1, 2)$, $c_1 = [4, 7)$, $c_2 = [0, 2)$, and $c_3 = [1, 2)$, and five buffers $b_0 = [0, 7)$, $b_1 = [5, 6)$, $b_2 = [5, 7)$, $b_3 = [1, 2)$, and $b_4 = [2, 6)$. Complete event-column enumeration has LP value four. One fractional solution uses weights $1/2, 1, 1/2, 1/2, 1/2, 1/2$ on

$$\{c_0, c_2\}, \{c_1, b_1\}, \{c_2, b_3\}, \{c_3, b_3\}, \{c_0, b_0, b_4\}, \{c_3, b_0, b_4\}.$$

Every core has total coverage one and each buffer has use at most one. Complete integer enumeration gives value three, attained for example by $\{c_0, c_2\}$, $\{c_3, b_0, b_1\}$, and $\{c_1, b_4\}$. The branch-and-price implementation closes this witness in 17 processed nodes, using both buffer-usage and pair-co-membership branching.

E Controlled-Truth Generator and Evaluation Contract

Each controlled instance has three core rows and eight candidate buffers. Core intervals are staggered. Every core receives a direct true member, and one or two cores receive a sequential member whose interval overlaps the direct member. Distractors are temporally plausible and have broad public outcome values. The generated truth has peak occupancy at most two; the same instances can therefore be evaluated under $C = 2, 3, 4$, where higher C enlarges only the analyst’s feasible set.

The composition benchmark reveals the true selected-buffer count q but never the selected identities. This isolates relation ambiguity from support-count error. The feasible point comparator maximizes an outcome-blind temporal score over all worlds with that q . A nearest- q rule ignores feasibility. Maximum-cardinality, maximum-overlap pair matching and positive-overlap connected components are support baselines that do not receive q .

For threshold η , the frontier certifies a positive decision when $\underline{H} \geq \eta$, a negative decision when $\overline{H} < \eta$, and ambiguity otherwise. If a feasible point world disagrees with the generated truth while the frontier is identified, two feasible worlds would lie on opposite sides despite both endpoints lying on one side, a contradiction. This explains the observed 100% ambiguity flag rate for feasible-point errors; the experiment measures how often such errors occur.

Candidate truncation ranks rows without using the public outcome. “True world representable” requires every true member to remain and the exact true partition to exist in the truncated master. “Frontier exists at true q ” only requires some alternative world with the same count. “Scalar covered” is weaker still: the true mean can fall between endpoints by coincidence even when the true joint world is absent. These nested validation concepts are reported separately.

F Artifact Claim Boundary

The repository contains deterministic small-instance oracles, exact fixed-time frontiers, numerical existential-time programs, branch-and-price status validators, controlled-truth benchmarks, public-data snapshot and count contracts, and aggregate evidence manifests. Supported claims are limited to the declared candidate universes and models. No artifact output should be interpreted as recovered partner identity, a complete private event, an estimate of realized capacity, or a statement about a city's proprietary production system.